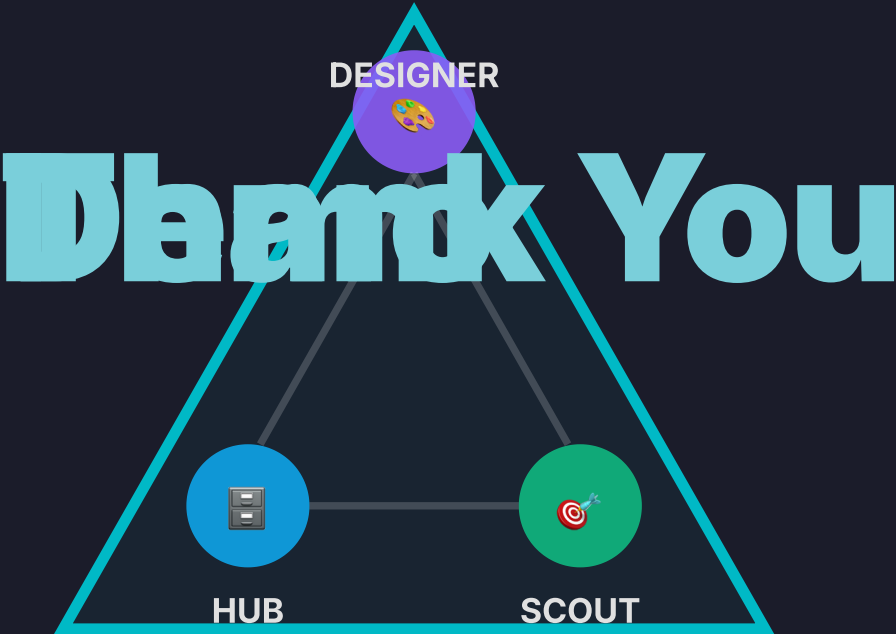




Team Details & Artefact links



Triangle Smart Targeting
& Real-Time Activation



TriStar

EPAM HACKATHON 2026 · LOYALTY AI PLATFORM

TEAM



Representative: *Ram Naresh, Kartikey Puri*

TEAM MEMBERS



Ram Naresh
AI Architecture & Backend

Rep



Kartikey Puri
Full Stack & Integration

Rep



Mohit Soni
Frontend & UX Design



Riyaj Shaikh
Backend & DevOps



Designer — AI Offer Copilot



Hub — Lifecycle State



Scout — Real-Time Activation

What we built: AI-powered three-layer loyalty engine — **Designer** turns plain-English objectives into risk-validated OfferBriefs; **Hub** tracks lifecycle in Redis; **Scout** activates at the perfect moment using GPS, weather & behaviour signals. Stack: **React 19 · FastAPI · Claude API · Redis · Azure.**



git.garage.epam.com/kartikey_puri/tristar
EPAM Garage · develop branch · Full source code





Problem Statement

Loyalty programs still behave like batch spreadsheets, not real-time brains — most offers are broad, not truly 1:1, and miss the moment that matters:


~70%

offers go unnoticed*


Hours

lost per campaign


Near 0

real-time context used

- High-value lapsed members still systematically under-targeted
- Most offers ignore real-time context — location, weather, behaviour
- No closed-loop feedback → same campaign mistakes repeated weekly
- Manual, error-prone workflows — mis-configs & margin losses slip through
- Siloed tools (LBM · AEM · SFMC · GL) — no lifecycle view draft-to-expiry



Batch Offers



Member Ignores



Revenue Lost

* Industry benchmark — open + click + redemption drop-off across typical loyalty campaigns

CATEGORY

AI-Powered Loyalty Personalisation

Aligned with Customer Experience & Digital Innovation



Solution

TriStar transforms loyalty into an **AI-powered, real-time engagement engine** — delivering the **right offer** to the **right member** at the **right moment**.

SYSTEM OVERVIEW — END-TO-END FLOW

LAYER 1 · DESIGNER

Biz Objective

Inventory Hint

Live Deal Feed

 Claude Sonnet 4.6

OfferBrief JSON

⬢ Risk Check

✓ Pass → Hub (draft)

✗ Critical → Block

↓ approved

THE HUB — SHARED CONTEXT STATE

Redis / In-Memory · draft → approved → active → expired · Audit Log · Expiry Sweep 300s

↓ context match

LAYER 3 · SCOUT — REAL-TIME ACTIVATION ENGINE

Flow A

Purchase Event

HMAC webhook · Determ. ≥ 70

Flow B

Hub Match

Claude AI > 60 · CASL dedup

Flow C

Partner Trigger

Claude Haiku · 24h brief

Flow D

Self-Activation

Customer-accept JWT



 Push Notification

 Queue for Later (8am)

 Activation Log

Tech Stack

Next.js 15

FastAPI

Claude Sonnet 4.6

Redis

Azure

Pydantic v2

Impact

+25%

lapsed member reactivation

80%

faster campaign creation

<200ms

real-time activation

Benefits

80%
Faster
Campaigns

+25%
Member
Reactivation

100%
Audit
Ready

For Marketers

- Generate validated OfferBriefs in seconds — not hours
- Built-in risk detection catches over-discounting before launch
- Inventory-driven recommendations increase campaign ROI
- Real-time risk flags surfaced before any offer goes live

For Members

- Offers arrive only when GPS, time & weather align
- Fewer, more relevant notifications — less fatigue
- Seamless, personalised loyalty experience at scale

For the Business

- Full lifecycle visibility: draft → approved → active → expired
- Immutable audit trail — OWASP & compliance ready
- Horizontal scale on Azure · millions of members

Future Enhancements

PRODUCT ROADMAP



Phase 1
MVP
Now



Phase 2
Intelligence
Q3 2026



Phase 3
Scale & Reach
Q1 2027

● Phase 1 — MVP ✓

Now · Hackathon

- ▶ Designer + Hub + Scout core
- ▶ Claude API brief generation
- ▶ 4-pattern risk detection
- ▶ Real-time context scoring
- ▶ JWT auth + Redis + Azure

✓ Live Now

- ✓ 3-layer AI engine
- ✓ Risk detection
- ✓ Real-time scoring
- ✓ Redis + Azure

● Phase 2 — Intelligence

Q3 2026

- ▶ ML-based segment targeting
- ▶ A/B testing framework
- ▶ Bilingual AI-generated offers
- ▶ Live analytics dashboard

🧠 Next Up

- ✓ ML targeting
- ✓ A/B testing
- ✓ Bilingual offers
- ✓ Analytics dash

● Phase 3 — Scale & Reach

Q1 2027

- ▶ Partner ecosystem open APIs
- ▶ Event-driven low-latency arch
- ▶ Mobile SDK + geofencing
- ▶ Multi-agent autonomous campaigns

🚀 Future

- ✓ Partner APIs
- ✓ Mobile SDK
- ✓ Geofencing
- ✓ Multi-agent AI

Built-in Guardrails (Phase 1)

🔒 1 offer / member / hr

🔒 No duplicates / 24h

🔒 Quiet hrs 10 pm–8 am

🔒 Critical risk → hard block

Security: OWASP Top 10 mitigations built in — parameterised queries, Zod + Pydantic input validation, JWT 1h expiry, Azure Key Vault for secrets, PII-free logs (member_id only).

